

19: Retrieving the data with PreparedStatement

To retrieve data from the database using a PreparedStatement, executeQuery() method will be used.

Steps for Retrieving the Data with Prepared Statement:

1. **Select Query with Placeholders:** Create a SQL SELECT query with placeholders (?) for dynamic values.

```
String sql = "SELECT id, sname, sage, scity FROM studentinfo WHERE  
id=?";  
  
pstmt = connect.prepareStatement(sql);
```

2. **Take User Input:** Collect the necessary input from the user to determine which data to retrieve.

```
Scanner scan = new Scanner(System.in);  
  
System.out.println("Kindly enter your id for which Data to be Retrieved: ");  
  
Integer id = scan.nextInt();
```

3. **Set the User Input:** Use the appropriate set of methods to inject the user-provided input into the placeholders.

```
pstmt.setInt(1, id);
```

4. **Use executeQuery():** Execute the query using the executeQuery() method of PreparedStatement. The result of the query is stored in a ResultSet object.

```
rs = pstmt.executeQuery();
```

- 5. Process the Retrieved Data:** Check if the ResultSet contains data, and then iterate through the result set to process or display the data. Use next() method to iterate through the records.

```
if (!rs.next()) {  
    System.out.println("No data found for the given ID.");  
} else {  
    do {  
        System.out.println("ID: " + rs.getInt("id"));  
        System.out.println("Name: " + rs.getString("sname"));  
        System.out.println("Age: " + rs.getInt("sage"));  
        System.out.println("City: " + rs.getString("scity"));  
    } while (rs.next());  
}
```

Code:

```
public class RetrieveStudentInfo {  
    public static void main(String[] args) {  
        String url = "jdbc:mysql://localhost:3306/db_name";  
        String username = "root";  
        String password = "password";  
  
        String sql = "SELECT id, sname, sage, scity FROM studentinfo WHERE  
id=?";  
  
        try (
```

```

        Connection connect = DriverManager.getConnection(url, username,
password);
        PreparedStatement pstmt = connect.prepareStatement(sql);
        Scanner scan = new Scanner(System.in);
    ) {
        System.out.print("Kindly enter your ID for which data to be retrieved: ");
        int id = scan.nextInt();

        pstmt.setInt(1, id);

        ResultSet rs = pstmt.executeQuery();

        if (!rs.next()) {
            System.out.println("No data found for the given ID.");
        } else {
            do {
                System.out.println("ID: " + rs.getInt("id"));
                System.out.println("Name: " + rs.getString("sname"));
                System.out.println("Age: " + rs.getInt("sage"));
                System.out.println("City: " + rs.getString("scity"));
            } while (rs.next());
        }

    } catch (SQLException e) {
        e.printStackTrace();
    }
}

```